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Research paper



Mechanical characteristics of spinal cord tissue by indentation

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ABSTRACT

The mechanical properties of brain and spinal cord tissue have proven to be extremely complex and difficult to assess. Due to the heterogeneous and ultra-soft nature of the tissue, the available literature shows a large variance in mechanical parameters derived from experiments. In this study, we performed a series of indentation experiments to systematically investigate the mechanical properties of porcine spinal cord tissue in terms of their sensitivity to indentation tip diameter, loading rate, holding time, ambient temperature along with cyclic and oscillatory dynamic loading. Our results show that spinal cord white matter tissue is more compliant than grey matter tissue with apparent moduli of 128.7 and 403.8 Pa, respectively. They show similar viscoelastic behavior with stress relaxation time constants of $\tau_1 = 1.38\,s$ and $\tau_2 = 36.29\,s$ for grey matter and $\tau_1 = 1.46\,s$ and $\tau_2 = 46.10\,s$ for white matter, while the initial peak force decreased by 54 % for grey and 59 % for white matter tissue. An increase of the applied loading rate by two orders of magnitude led to an approximate doubling of the apparent modulus for both tissue types. Thermal variations showed a decrease in apparent modulus of up to 30 % after heating from 20 to 37.0 °C. Our dynamic tests revealed a significant influence of cyclic preload on the stiffness, with a drop of up to 20 % and a relative decrease of up to 60 % after the first cycle compared to the total modulus drop after five cycles for spinal cord grey matter tissue. Oscillatory indentation experiments identified similar loss moduli for spinal cord grey and white matter tissue and a higher storage modulus for white matter tissue. This work provides systematic insights into the mechanical properties of spinal cord tissue under different loading scenarios using nanoindentation.

1. Introduction

The spinal cord is essential for the function of the central nervous system, yet its mechanical properties remain relatively poorly understood. Despite the considerable experimental efforts in recent years that have focused intensively on investigating the mechanical properties of grey and white matter central nervous system tissue (Chatelin et al., 2010), only a subset of these experiments has targeted the mechanical characterization of spinal cord tissue (Cheng et al., 2008). Overall, all studies show a variety of partly contradictory results, which can be attributed to the great inconsistency in sample preparation, test protocols, and environmental conditions (Budday et al., 2020). Sample preparation alone can cause a difference of several orders of magnitude in the derived stiffness, depending on whether the dura mater has been removed or not (Bilston and Thibault, 1995; Karimi et al., 2017).

Understanding and accurately quantifying the mechanical properties of spinal cord tissue has important implications for clinical and scientific problem, as mechanics plays an essential role for various physiological and pathological conditions, which is increasingly recognized (Kuhl, 2018). Spinal cord disorders, such as syringomyelia or spinal tumours as well as stenosis of the spinal canal, are partially mechanically driven, as the compression of spinal cord tissue causes neurological dysfunction (Bilston, 2011). Mechanical factors also play a decisive role in guiding axonal (re)growth (Koser et al., 2015), multiple sclerosis, and spinal cord injuries (Franze et al., 2013). In the context of spinal cord diseases, especially the search for therapeutic interventions for spinal cord injury has been one of the most prominent research topics. Most of these efforts have concentrated on molecular processes and physical parameters, such as electricity, magnetism, and ultrasound (Hu et al., 2023). However, only a few studies have investigated mechanical properties. Those for instance revealed that spinal cord tissue stiffens in regenerative species (Möllmert et al., 2020) but becomes softer in non-regenerative species (Moeendarbary et al., 2017) after severe injury.

A better understanding of the mechanical properties of spinal cord tissue in homeostasis and pathology will stimulate further research to

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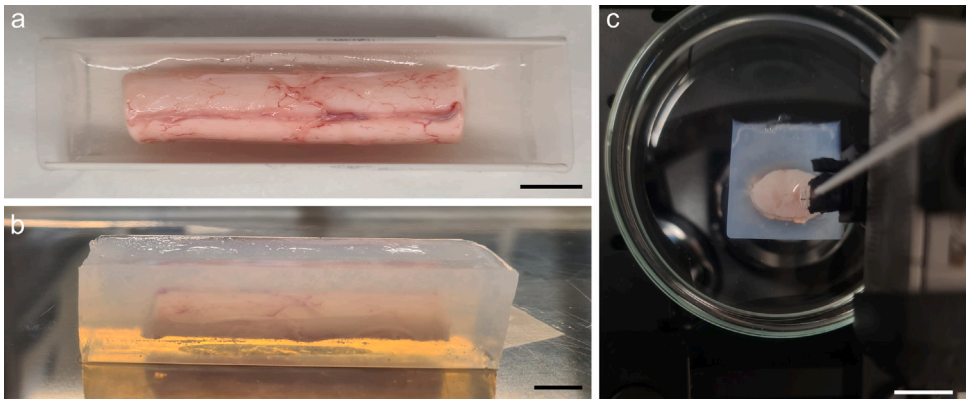


Fig. 1. Sample preparation process. (a) Exposed freshly harvested spinal cord segment in mould on 3–5 mm agarose bed. (b) The same segment covered with agarose. (c) Experimental setup where a 7 mm-thick spinal cord slice of freshly harvested spinal cord tissue is placed in a 6 cm-diameter petri dish and mounted underneath a Chiaro Nanoindenter^{TM}. The spherical tip is located at the bottom of the dark gold-plated cantilever. Scale bars denote 1 cm.

couple mechanical modeling with molecular biology and biophysics to simulate processes like neuroregeneration and spinal cord remodeling after spinal cord injury. This will provide us with the necessary expertise to develop and test therapeutic strategies to establish new treatments, e.g., by improving the locomotion recovery in mammals after spinal cord injury (Wang et al., 2022). The development of mathematical continuum models that enable the simulation of these biomechanical processes can be a valuable tool to speed up this process. However, the accuracy and usefulness of those models depends to a large extent on a thorough investigation of the mechanical properties of spinal cord tissue in different temporal and spatial domains along with different regimes of deformation (Budday, 2023).

Experimental efforts to characterize mechanical properties of spinal cord tissue revealed considerable discrepancies due to different length and time scales, tissue preparation, experimental setup, preconditioning, and environmental conditions (Cheng et al., 2008; Chatelin et al., 2010). Specimen preparation, with particular attention to the removal or non-removal of the dura mater, plays a crucial role in the overall mechanical response and must be carefully considered when comparing results (Cheng et al., 2008), as the dura mater is much stiffer than spinal cord tissue (Zhang et al., 2016; Jin et al., 2023). Additionally, in order to properly investigate the mechanical properties of nervous tissue in general, it is essential to understand the mechanical differences between grey and white matter. In this context, the results of indentation experiments reported in the literature differ considerably for spinal cord tissue. Koser et al. (2015), Möllmert et al. (2020) and Moeendarbary et al. (2017) state that spinal cord grey matter tissue is stiffer than white matter tissue. Contradictory to these findings, Cooper et al. (2020) report no difference between the stiffness of spinal cord grey and white matter tissue. Interestingly, Van Dommelen et al. (2010), Budday et al. (2015) and Kaster et al. (2011) observed that white matter is stiffer than grey matter in bovine and porcine brain tissue. It is worth noting that within these studies, the indenter dimensions range from μm to mm with either flat-punch or spherical indenters.

To analyze these inconsistencies, we employ a method at the intermediate scale that is robust, dependable and reproducible to provide further insights into the mechanical properties of spinal cord grey and white matter tissue. For this purpose, we systematically investigate possible influences of the spherical indenter radius, loading rate, holding time, temperature and buffer submersion time by performing a series of 2960 indentation tests. In addition, we perform a grid scan of 1200 indentation points on a single spinal cord slice to quantify the local heterogeneity of spinal cord tissue stiffness.

2. Methods

2.1. Sample preparation

We performed our experiments on freshly harvested porcine spinal cord of three to four months old female pigs, which were provided by a local slaughterhouse (Contifleisch GmbH und Unifleisch GmbH, Erlangen, DE). This involved some uncertainties regarding the handling and state of the samples, but allowed us to avoid animal or human studies. Due to the extremely soft nature of spinal cord tissue and its tubular geometry, dissecting and cutting samples without structural support can introduce unintentional mechanical preconditioning, which might alter the mechanical response of the tissue during the subsequent indentation experiments. Moreover, cutting suitable samples from an unsupported spinal cord leads to a rather uneven surface topology that significantly impairs indentation testings. To address these challenges, we extracted 7 cm long segments from the porcine spinal cord and embedded them in a block of agarose to improve the cuttability of the tissue (Iwashita et al., 2014), as illustrated in Fig. 1. The agarose frame around the spinal cord tissue provides structural support when the tissue is cut into suitable 7 mm thick slices, but has no influence on the mechanical properties of spinal cord tissue (Iwashita et al., 2014). We hydrated the spinal cord segments with Dulbeccos phosphate buffered saline solution before embedding them into agarose and kept the tissue refrigerated at 4 °C as long as it was not being tested. This procedure slows down the degradation of the tissue and prevents tissue dehydration. All tests were performed within 33.5 h *post mortem* (see Tab. S1).

2.2. Experimental setup

We glued the agarose frame of the specimen on the bottom of the petri dish and submerged the specimen in Dulbeccos phosphate buffered saline solution (DPBS) during the experiments. The petri dish with the fixed specimen was placed in a holder under a Chiaro Nanoindenter^{TM} (Optics11 Life, Amsterdam, NL). In addition, we utilize a heating system for Petri dishes (Pecon GmbH, Erbach, DE) to control the temperature during the tests if necessary. For the indentation experiments, we used spherical indenters with a cantilever stiffness of 0.5 N/m and radii ranging from 50 μm to 250 μm. Due to manufacturing limitations, there may be slight deviations in the final tip radius (the ranges are as follows: 250 ± 25 μm, 100 ± 10 μm, 50 ± 10 μm). Fig. 1c illustrates a spinal cord slice in an agarose frame during indentation.

The Chiaro Nanoindenter^{TM} uses a piezo-electric translator to move the tip vertically during indentation and calculates the force of the

tissue response by measuring the bending of the cantilever on which the indenter tip is located. The bending is calculated using an interferometer, which records the phase shift between light that is reflected at different interfaces of the indenter probe. The light is reflected by the optical fiber-to-gap interface and the gap-to-cantilever interface (Chavan et al., 2012).

2.3. Indentation procedure

All indentations were performed under indentation control and employing trapezoidal loading-holding-unloading profiles. With the exception of the experiments on temperature sensitivity and the investigations on the sensitivity of the measured properties in relation to the time the sample was immersed in a buffer, all tests were conducted at room temperature. In indentation controlled experiments, a predefined threshold value regulates the detection of the sample surface during indentation. This surface detection is evaluated in real-time based on the cantilever bending. It provides the internal software of the testing device a reference value from which it can apply the desired indentation depth.

Initially, we investigated possible effects of submersion duration in distilled water and Dulbeccos phosphate buffered saline solution (DPBS) on the measured tissue properties to ensure consistent testing conditions. To this end, we tested 20 samples extracted from six porcine spinal cords. On each sample, we performed a grid of 20 indentation points with a loading rate of 50 μm/s and a maximum indentation depth of 50 μm. This test was repeated four times at the same location on the sample and at different time points during the submersion period. We performed the first scan right after mounting the sample, followed by a waiting period of five minutes. Then, we repeated the grid scan, waited ten minutes, repeated the scan again, waited 30 min, and repeated the scan one last time. The sketch in Fig. S1 shows a summary of the procedure for an exemplary curve. To compare possible effects of submersion duration, we then computed the apparent modulus of all indentation points on the grid for all time points. We normalized each set of four apparent moduli for each individual location (corresponding to the course of stiffness of this location over the period of submersion time investigated here) by dividing the set by the maximum value of the set. Finally, we calculated the mean and standard deviation of all normalized values for each time point to quantify the influence of submersion duration on the tissue stiffness. We conducted this study at 37 °C body temperature (the maximum temperature investigated in this study), as we expect that tissue degradation and other biochemical effects are most pronounced at high temperatures.

Additionally we performed a grid scan of 253 indentations with a depth of 50 μm, using spherical tips with radii of 50 μm, 100 μm, and 250 μm. Subsequently, all indentations employed the 100 μm-radius indenter. To examine the impact of the loading rate, we conducted nine indentations in spinal cord grey and white matter tissue, respectively, at rates of 1 μm/s, 5 μm/s, 10 μm/s, 15 μm/s, 20 μm/s, 30 μm/s, 50 μm/s, 100 μm/s, and 150 μm/s. Additionally, to elucidate the influence of the holding time, eight indentations were performed on grey matter tissue, incorporating holding times of 1 s, 2.5 s, 5 s, 10 s, 20 s, 40 s, 80 s, and 160 s.

This series was supplemented by 12 stress relaxation experiments, featuring a loading and unloading rate of 30 μm/s and an extended holding time of 600 s. These experiments were conducted at six distinct locations in grey and white matter, respectively. To normalize the stress relaxation data, we divided each value of the force data by the maximum value in the set, which corresponds to the first value in the set.

To further assess the viscoelastic properties of spinal cord tissue, we performed six cyclic indentation experiments, each with five loading–unloading cycles, as well as eight dynamic mechanical analysis measurements at different indentation depths of 10, 20, 30, and 40 μm, an oscillatory amplitude of 1 μm, and frequencies of 0.5, 1, 2, 4, 10 and 20 Hz. Both of these analyses were performed with a loading rate of 10 μm/s. To normalize the cyclic loading data, we divided each set of peak forces and apparent moduli respectively for each individual location by the maximum value in the set, which corresponds to the first value in the set.

Complementary, we investigated the temperature sensitivity of the mechanical properties of spinal cord white matter tissue through heating up and cooling down a sample and testing it at different temperatures between 20° and 37.5 °C. We were able to collect experimental results from 14 measurements during the cooling process and eight measurements during the heating process within the specified temperature interval. To additionally investigate the influence of temperature on the stiffness of spinal cord grey and white matter tissue (and the corresponding ratio), we conducted a larger study involving 549 indentations on 18 samples from three different porcine spinal cords. We performed a grid of four indentations on the same area in spinal cord white and on the same area in spinal grey matter tissue of each sample at different temperatures in the range of 20° to 37.0 °C with slight variations of ± 0.16 to ± 0.38 °C between the different samples due to experimental conditions (for this experiment, we started at room temperature and heated up the samples). We then calculated the mean of the apparent moduli for each grid and divided the set of mean apparent moduli of each tissue type by the first value in the set (which corresponds to the mean stiffness of that grid at the lowest temperature). Similar to the investigation of the influence of immersion time, we then calculated the mean value of the calculated mean apparent moduli of the individual grids at the investigated temperature levels. Additionally, we computed the ratio between the mean apparent moduli of spinal cord grey and white matter tissue for each temperature.

Finally, we also conducted a series of 1200 indentations over a full spinal cord slice, with a loading rate of 30 μm/s and without holding time to assess how properties change over the different areas of the spinal cord slice.

In total, this investigation comprised a series of 4160 indentations on 45 samples of 16 porcine spinal cords from 16 different pigs, 1600 to compare the influence of submersion periods in DPBS and distilled water, 759 to assess sensitivity relative to punch diameter, 18 to evaluate sensitivity with respect to loading rate, 20 to analyze sensitivity concerning holding time, 549 to investigate the influence of the ambient temperature, 14 for dynamic analyses, and 1200 indentations for measurements of a whole spinal cord slice.

When we performed multiple measurements at the same location, we maintained a minimum interval of 1 min between two consecutive indentations to avoid possible effects of preconditioning. Table 1 summarizes the experimental setups for all the tests described above.

2.4. Indentation data analysis

Fig. 2 shows a typical force–displacement curve, highlighting the individual phases during an indentation experiment. The segmentation of this curve into approach, loading, holding and unloading for all subsequent data visualizations is based on the contact point identified by the internal software of the Chiaro Nanoindenter™, which is explained in Section 2.3. The point with the highest measured force determines the end of the loading phase and the beginning of the holding phase. As the duration of holding is user-defined, this information can be used to determine the end of the holding phase and the unloading phase.

We used a custom-written image analysis tool to allocate indentation points to either grey or white matter. To this end, two images are correlated: the first is an image of the spinal cord sample, as shown in Fig. 3a; the second image shows the indenter tip right above the first indentation location of a matrix scan, as illustrated in Fig. 3b. Due to the gold-plated top of the cantilever (which increases its reflectivity), the position of the cantilever above the sample and thus the position of

Table 1
Summary of experimental conditions for the mechanical tests of this study.

Test	# Indents	# Samples	# Animals	Max. depth (μm)	Rate (μm/s)	Holding time (s)	Tip radius (μm)
Buffer submersion sensitivity	1600	20	6	50	50	1	97.5 and 102.5
Radius comparison	759	1	1	50	30	1	Varying
Loading rate sensitivity	18	1	1	50	Varying	10	98
Holding time sensitivity	8	1	1	50	30	Varying	98
Stress relaxation	12	1	1	50	30	600	102 and 110
Cyclic loading	6	1	1	50	10	0	110
Dynamic mechanical analysis	8	1	1	Varying	–	–	110
Thermomechanical analysis	549	18	3	50	50	1	97.5
Heterogeneity analysis	1200	1	1	50	30	10	102

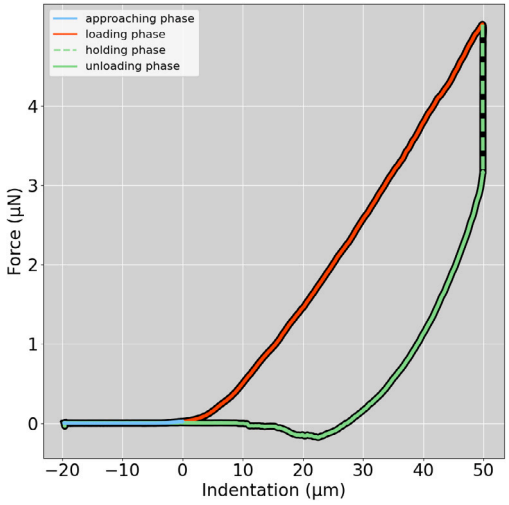


Fig. 2. Typical force–displacement curve for a spinal cord indentation experiment in black with the approach, loading, holding, and retraction phases highlighted in blue, red, green dashed, and green solid, respectively.

the spherical indenter in relation to the tissue can be easily determined. We calculate the position of the start point of the grid on the reference picture as follows: One edge of the rectangular shape at the bottom of the petri dish is used as a fixed reference length for all experiments. The vector indicating the start position of the grid indentation is transformed to the reference image using the equation $\boldsymbol{v}_S^* = f_S \cdot \boldsymbol{v}_S$. The transformation factor f_S is the ratio between the edge length in the reference image and the edge length in the start image (in pixel per mm). Fig. 3 illustrates this routine for an exemplary spinal cord slice. Fig. 3c shows the resulting grid of indentation points on the spinal cord slice, which is then used to assign the individual indentations to either grey or white matter manually. We mount the camera above the sample and control it wirelessly to ensure the same viewing angle for all images and to minimize possible deviations of the derived start point compared to the real start point due to camera-related viewing angle distortions. We have quantified possible distortions for the visual allocation method described above caused by submerging the sample in DPBS to less than 10 μm, which is tolerable for the spatial scale investigated in this study (see Fig. S2).

Alternatively, a grid can be selected that covers the entire transverse surface of the spinal cord slice. This enables the allocation of indentation points by simply comparing a map-plot of results with the reference image and deducing which points were on grey and which were on white matter. This approach is illustrated in Fig. 17. We assign the indentation points at the boundary between the grey and white matter tissue of the spinal cord based on the percentage of area of the corresponding square in grey or white matter tissue. Fig. S3 shows a visualization of this procedure. For single indentation experiments, we visually determine if it was in grey or white matter tissue.

2.5. Mechanical properties

In a first approximation, we employ contact mechanics based on Hertz's theory to obtain an apparent modulus from the loading segment of each indent. The findings in Lin et al. (2009) show that the classical Hertz model for a parabolic indenter (Sneddon, 1965) is only suitable for indentations up to a ratio of 0.16 between the maximum indentation depth and the radius of the indenter. Therefore, we utilized a modified version of the Hertz model, the solution to an axisymmetric frictionless contact problem, also called the Boussinesq's problem (Sneddon, 1948, 1965), specifically for spherical indenters, which was first derived by Segedin (1957). In reference to Hertz's groundwork (Hertz, 1882), we will refer to this model as the Hertz model in the following. We fit the loading phase of the experimentally generated force–displacement curves to the following relation

$$F = \frac{E_{\text{app}}}{1-\nu^2}\left[\frac{R^2+a^2}{2}\ln\left(\frac{R+a}{R-a}\right) - aR\right]$$
$$d = \frac{a}{2}\ln\left(\frac{R+a}{R-a}\right), \tag{1}$$

where R denotes the radius of the indenter, F is the measured force, d the displacement and a the radius of the circular contact area between indenter and sample. All fits of the so far discussed model were processed with an assumed Poisson's ratio $\nu = 0.5$ and a custom written Python code that employs the *lmfit* function for non-linear least-squares minimization. The apparent modulus, E_{app}, which is derived by this method is denoted as "apparent" to emphasize that the Hertz model assumes linear elasticity. This does not correspond to the typically observed viscous and porous behavior of brain tissue (Chatelin et al., 2010; Greiner et al., 2021). Therefore, in the following, the apparent modulus is considered as an approximation of properties under very specific loading conditions in order to rather quantify ratios or trends. It should not be equated with more meaningful mechanical properties derived by nonlinear continuum mechanics, which result from inverse parameter identifications (Hinrichsen et al., 2022).

To derive averaged apparent moduli for spinal cord tissue, we overlaid a heat map of the moduli for a 40 by 30 indentation grid on a whole spinal cord slice with an image of the corresponding slice and assigned individual indentation points to white and grey matter accordingly, see Fig. 17.

We analyze stress relaxation experiments by fitting a multi-term Prony series

$$\sigma(t) = \sigma_\infty + [\sigma_0 - \sigma_\infty]\sum_{i=1}^{n}\exp(-t/\tau_i) \tag{2}$$

with n characteristic time constants τ_i and the plateau stress σ_∞ at $t \to \infty$. For this purpose, we have written a custom Matlab code that uses the function *fmincon*, which applies the method of interior points to find the minimum of constrained nonlinear functions.

For dynamic mechanical analysis, the post-processor of the testing machine was used to derive the complex modulus of elasticity $E^* = E' + iE''$. The frequency-dependent storage and loss modulus, E' and

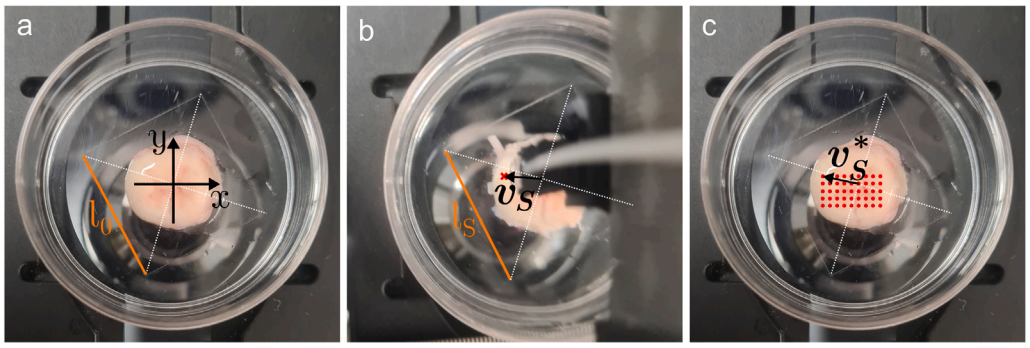


Fig. 3. Visualization of indentation point identification routine to allocate individual experiments to either grey or white matter. (a) Reference picture of the mounted spinal cord slice with reference length l_0 and coordinate origin in the center of the reference frame. (b) Starting point of the indentation grid with vector $\boldsymbol{v}_S$ from origin to start point and current reference length l_S. (c) Derived indentation grid in red with corresponding start point vector in reference coordinate system $\boldsymbol{v}_S^*$.

E'' respectively, are derived by fitting the following equations to the experimental data

$$\begin{aligned} E'(f) &= \frac{1}{2}\frac{F_0(f)}{d_0}\cdot\cos\delta\cdot\frac{[1-\nu^2]}{\sqrt{dR}} \\ E''(f) &= \frac{1}{2}\frac{F_0(f)}{d_0}\cdot\sin\delta\cdot\frac{[1-\nu^2]}{\sqrt{dR}} \\ \tan\delta &= \frac{E''}{E'} \end{aligned} \tag{3}$$

where f denotes the cyclic motion frequency, d_0 the amplitude of indentation depth, F_0 the measured force, and δ the phase lag between oscillations of indentation and load.

Furthermore, we employ an inverse parameter identification routine from Hinrichsen et al. (2022) which uses a finite element (FE) model implementing the finite viscoelastic model described in Reese and Govindjee (1998), Budday et al. (2017b) and Hinrichsen et al. (2024) that is based on a multiplicative split of the deformation gradient $\boldsymbol{F}$ into an elastic and a viscous contribution. The deformation gradient can be obtained as the gradient $\nabla_{\boldsymbol{X}}\boldsymbol{\varphi}(\mathbf{X},t)$ of the deformation $\boldsymbol{\varphi}(\boldsymbol{X},t)$, which maps from the material configuration $\boldsymbol{X}$ to the deformed configuration $\boldsymbol{x}$ at time t. The multiplicative split is then written as

$$\boldsymbol{F} = \boldsymbol{F}^{\mathrm{e}}\cdot\boldsymbol{F}^{\mathrm{v}}, \tag{4}$$

where the superscripts e and v denote the elastic and viscous contributions, respectively. Furthermore, we can write the viscoelastic strain energy as sum of an equilibrium and non equilibrium parts, where the latter contains the contributions of the generalized Maxwell element(s)

$$\psi = \psi^{\mathrm{eq}} + \psi^{\mathrm{neq}} \tag{5}$$

Each spring is described by a compressible one-term Ogden model (Ogden, 1972; Holzapfel, 2002; Hinrichsen et al., 2022) defined by the strain energy function

$$\Psi = \Psi_{\mathrm{iso}} + \Psi_{\mathrm{vol}} \tag{6}$$

with

$$\Psi_{\mathrm{iso}} = \frac{2\mu}{\alpha^2}\left[\bar{\lambda}_1^{\alpha} + \bar{\lambda}_2^{\alpha} + \bar{\lambda}_3^{\alpha} - 3\right] \tag{7}$$

and

$$\Psi_{\mathrm{vol}} = \frac{\kappa}{4}\left[J^2 - 1 - 2\ln(J)\right] \tag{8}$$

where $\bar{\lambda}_a = J^{-1/3}\lambda_a$ are the isochoric principal stretches and $J = \det\boldsymbol{F}$ denotes the volume ratio. The one-term Ogden model was shown to well capture the mechanical behavior of human brain tissue, especially the pronounced compression-tension asymmetry (Budday et al., 2017b; Hinrichsen et al., 2022; Budday et al., 2017c). Hence, each spring is characterized by the shear modulus μ—we adapt the modified formulation in Budday et al. (2017a)—and the nonlinearity parameter α. The bulk modulus κ characterizes the compressibility in both springs but is not a free parameter, as we calculate it from the shear modulus μ using the relation

$$\kappa = \mu\frac{2[1+\nu]}{3[1-2\nu]} \tag{9}$$

from the linear regime, where we use the Poisson's ratio $\nu = 0.45$. Due to known numerical limitations, we explicitly do not model an incompressible material with $\nu = 0.5$ in our inverse parameter identification routine. In our previous work (Hinrichsen et al., 2022), we have identified hyperelastic parameters for Poisson's ratio values $\nu = \{0.45, 0.49\}$ with no significant differences in the obtained fitting error. Still, the identified parameters will differ with higher shear moduli and nonlinearity parameter for lower Poisson's ratios. Since we assume purely deviatoric contributions to the viscosity, this introduces one additional parameter η (Budday et al., 2017c). The rheological representation of the model is shown in Fig. 4. We denote the parameters of the equilibrium spring and the viscoelastic Maxwell element with the subscripts ∞ and 1, respectively. Subsequently, the set of parameters that needs to be determined includes μ_∞, μ_1, α_∞, α_1, and η_1. We implement the model in C++, using the finite element library *deal.ii* (Arndt et al., 2021).

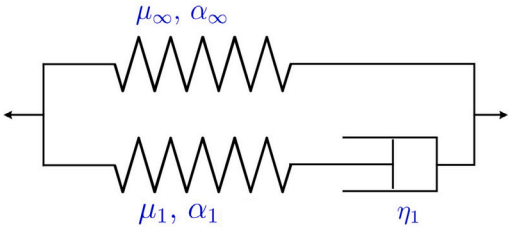


Fig. 4. Rheological scheme of the implemented generalized Maxwell model with one Maxwell element. Each nonlinear spring represents a hyperelastic one-term Ogden model characterized by the shear modulus μ and the nonlinearity parameter α. η denotes the viscosity of the dashpot element.

3. Results

3.1. General indentation characteristics

A typical force–displacement curve from an indentation test consists of three phases: an approach phase, in which the indenter is lowered until it reaches the surface of the tissue (a plateau at $F = 0$ in the force–displacement diagram, e.g. Fig. 2), a loading phase, which begins when an increase in force is measured, and a retraction or unloading phase, in which the indentation depth and thus the force gradually decreases. The retraction phase can also be initiated by a predefined holding phase in order to investigate the time-dependent mechanical properties of the spinal cord tissue. At a maximum indentation depth of 50 μm,

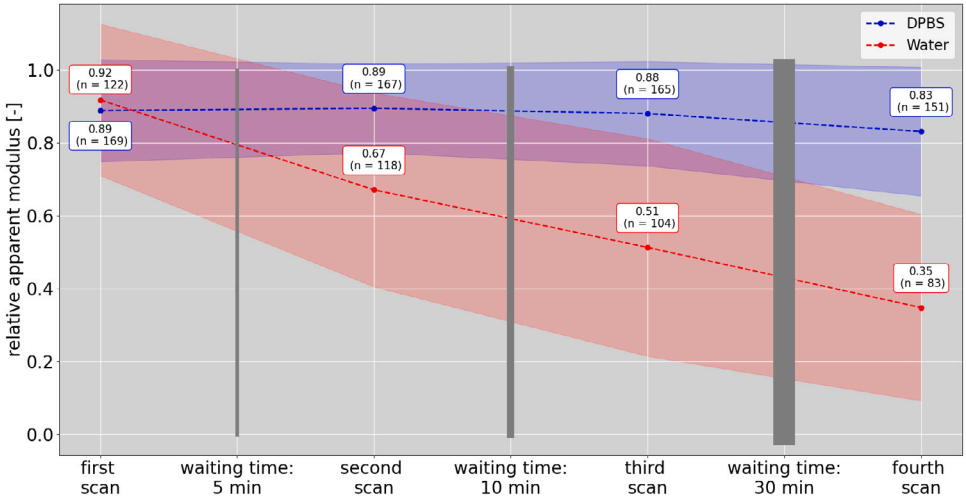


Fig. 5. Softening due to the duration of submersion in the buffer. The dashed blue and red lines show the trend in tissue stiffness over successive indentations when the sample was immersed in DPBS and distilled water, respectively. The light blue and light red areas visualize standard deviations. The text boxes show the mean relative apparent moduli of all indentation points together with the number n of indentations per experimental stage. Experimental conditions: 20 indentations per sample on 20 samples of six porcine spinal cords, maximum indentation depth of 50 μm, loading rate of 50 μm/s, holding time of 1 s, tip radii of 102.5 μm (16 samples of four different spinal cords) and 97.5 μm (four samples of two different spinal cords).

we held the indentation depth constant and the force decreased upon relaxation. As the unloading process neared its end, the recorded force briefly dipped below zero, revealing adhesive effects, before eventually returning to its initial value.

3.2. Sensitivity of the measured properties with respect to submersion time

In order to investigate possible effects of the duration a sample is submerged in a buffer we tested 20 samples from six porcine spinal cords. We tested ten samples immersed in DPBS and ten samples immersed in distilled water. Fig. 5 shows the trends of the relative apparent moduli for indentation experiments in DPBS and in distilled water over a time duration of approximately one hour. The trend of the indentations experiments on samples submerged in DPBS clearly shows a stable tendency with mean relative apparent moduli of 0.89, 0.89, 0.88 and 0.83 for scan 1 (initial test), 2 (testing time starting at 10 min), 3 (testing time starting at 25 min) and 4 (testing time starting at 60 min) respectively. On the contrary the trend of relative apparent moduli of experiments on samples submerged in distilled water shows a material softening with 0.92, 0.67, 0.51 and 0.35 for scan 1, 2, 3 and 4 respectively.

3.3. Sensitivity of the measured properties with respect to the tip radius

To test the reliability of the measuring system and our theoretical approach in Eq. (1), we performed three grid indentations with 253 indentation points using spherical indenters with radii of 49.5 μm, 105 μm, and 273 μm as well as a cantilever stiffness of 0.5 N/m each. Fig. 6 illustrates the results. On average the apparent moduli amount to 1267.7, 1298.5 and 936.6 Pa for the radii 49.5, 105 and 273 μm respectively. The black squares resemble insufficient measurements, where no apparent modulus could be derived. This was observed for two and 14 of the total 253 indentations during the tests with a radius of 49.5 and 273 μm, respectively. Preliminary experimental observations had shown that the indenter tip with a (desired) radius of 100 μm produced the most consistent results and was therefore tested first, followed by tests with an indentation radius of 49.5 and 273 μm. Approximately 75 min were required for one run of 253 indentations. This could explain that the greatest deviation from the overall mean value is observed for the last test run with an indenter of 273 μm. This gives rise to the assumption that the longer a spinal cord specimen is immersed in phosphate-buffered saline solution, the more compliant it becomes. Another reason for the larger difference could be the limitation of the Hertz contact model for the strain range in which we test our samples. The Hertz model assumes linear elasticity, which is a simplification of the actual situation during 50 μm indentation tests. As explained in the previous section, the derived apparent modulus is used to quantify ratios and trends and is not considered a mechanical parameter comparable to one derived by inverse fitting of material models to experimental measurement data. Due to the best consistency, all subsequent measurements are carried out with a desired indenter radius of 100 μm and a cantilever stiffness of 0.5 N/m.

3.4. Rate-dependent properties of spinal cord grey and white matter

The results of our loading rate sensitivity study are shown in Figs. 7 and 8. The figures illustrate the highly rate-dependent characteristics of spinal cord grey and white matter tissue on one exemplary sample. We performed a series of 18 indentation tests with nine different loading rates of 1 μm/s, 5 μm/s, 10 μm/s, 15 μm/s, 20 μm/s, 30 μm/s, 50 μm/s, 100 μm/s and 150 μm/s. Fig. 7 shows that with a displacement-controlled trapezoidal load/hold/unload profile, the indentation force gradually increased to its maximum value during loading, partially relaxed during holding, and finally fell to zero after unloading. The maximum indentation force increased significantly with increasing loading speed. Within the loading rates from 1 μm/s to 160 μm/s, the exemplary maximum force in the spinal cord white matter tissue increased from 1.02 μN to 1.88 μN and the exemplary maximum force in the spinal cord grey matter tissue increased from 2.24 μN to 4.61 μN. With an increase of the loading rate by two orders of magnitude, the maximum force in both grey and white matter approximately doubled.

Fig. 8 depicts the exemplary force-indentation behaviour of one sample of spinal cord grey and white matter tissue during the loading phase and the corresponding apparent moduli for different loading rates. The maximum forces and apparent moduli increased with increasing loading rate with the exception of the indentation with v = 100 μm/s on spinal cord white matter and the indentation with v = 20 μm/s on spinal cord grey matter tissue. These two discrepancies could either be attributed to the inadequate surface detection of the internal software or biological residues on the indenter tip. Apart from these two outliers, however, the basic trend of increasing stiffness with increasing loading speed is clearly observable. Within the range of

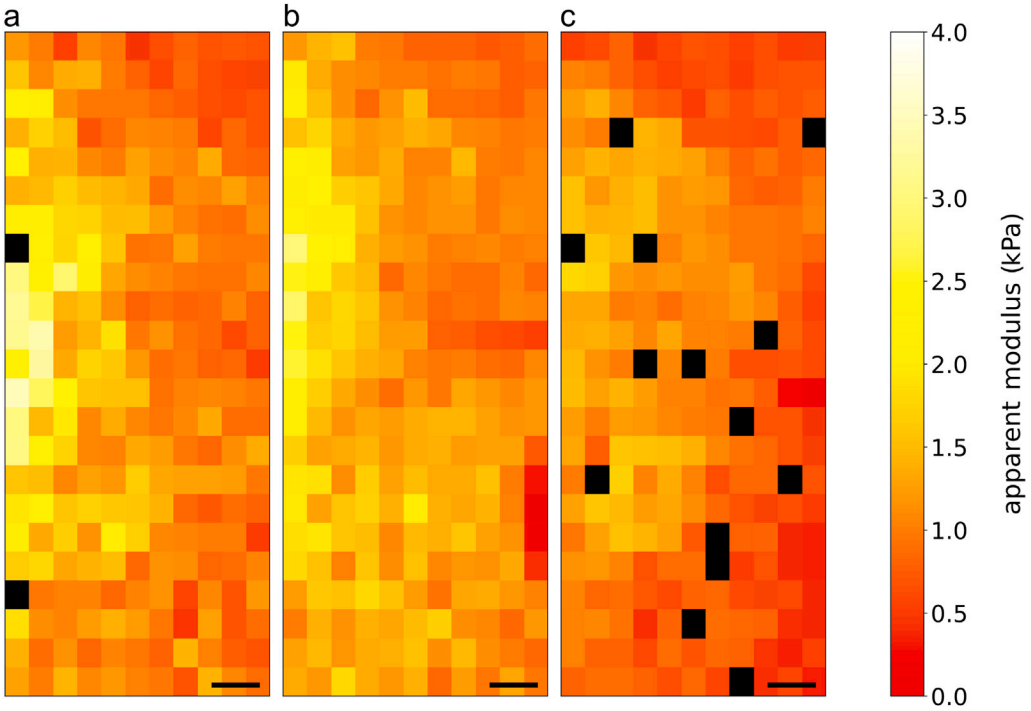


Fig. 6. Maps of the apparent modulus for a section on a spinal cord slice with tip radii of (a) 49.5, (b) 105 and (c) 273 μm, containing both spinal cord grey and white matter tissue. Black squares indicate faulty measurements. The results for a tip radius of 105 μm were recorded first, followed by 49.5 and 273 μm tips. Scale bars denote 400 μm. Experimental conditions: three times 253 indentations with different tip radii, respectively, on the same location in one sample, maximum indentation depth of 50 μm, loading rate of 30 μm/s, holding time of 1 s.

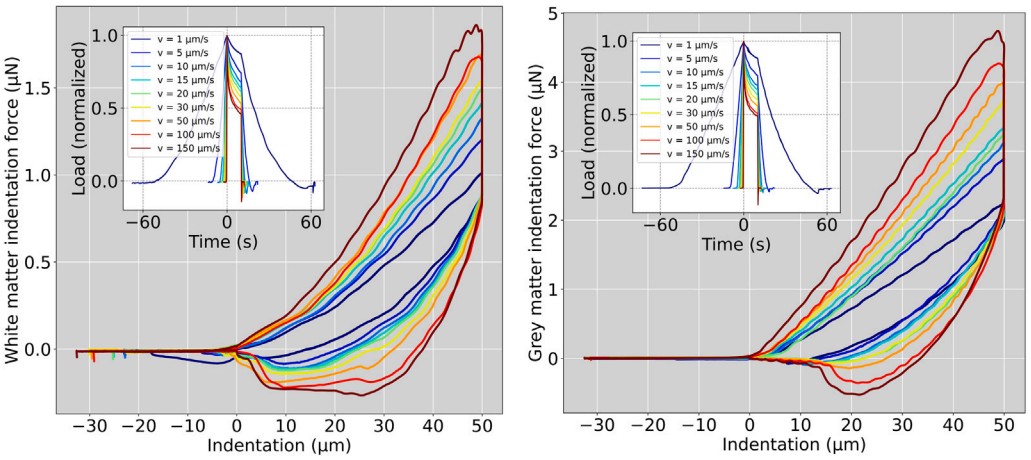


Fig. 7. Loading rate sensitivity of grey and spinal cord white matter tissue. The slope of the force-indentation curves and maximum reached force increases with increasing loading rate. The inset shows the corresponding stress relaxation behaviour over a holding period of 10 s. Experimental conditions: 18 indentations on two locations (nine indentations each) of one sample, maximum indentation depth of 50 μm, varying loading rate, holding time of 10 s, tip radius of 98 μm.

investigated loading rates from 1 μm/s to 150 μm/s, the exemplary apparent modulus of spinal cord white and grey matter tissue increased from 170.2 Pa to 335.3 Pa and from 405.7 Pa to 916.3 Pa, respectively.

To further quantify exemplary rate-dependent properties of spinal cord grey and white matter tissue, we identified parameters for the model defined in Section 2 through an inverse finite element analysis (Hinrichsen et al., 2022) based on the response during the loading segment of three of the nine loading rates performed: 1 μm/s, 50 μm/s, and 150 μm/s. The identification entailed an iterative process wherein finite element simulations are conducted with varying parameter values until a paremeter set is obtained for which the experimental and simulated indentation force–displacement curves match to a sufficient degree. Fig. 9 shows an example of the displacement field of a finite element analysis for 50 μm indentation depth and a spherical indenter with a radius of 98 μm.

The optimization routine was initialized with all parameters set to 1.0. Table 2 presents the optimized parameters for both spinal cord grey and white matter tissue. The parameters were subsequently employed in forward simulations to predict the tissue response across varying loading speeds, including those not used during parameter identification. We employed the coefficient of determination, denoted by R^2, to quantify the accuracy of the predictions.

As shown in Fig. 10 left, the model agrees well with the experimental observations for the spinal cord white matter tissue ($R^2 > 0.96$). The comparably low R^2 for 100 μm/s can be attributed to anomalies within the experimental data, with almost overlapping experimental force–displacement curves for velocities of 50 μm/s and 100 μm/s.

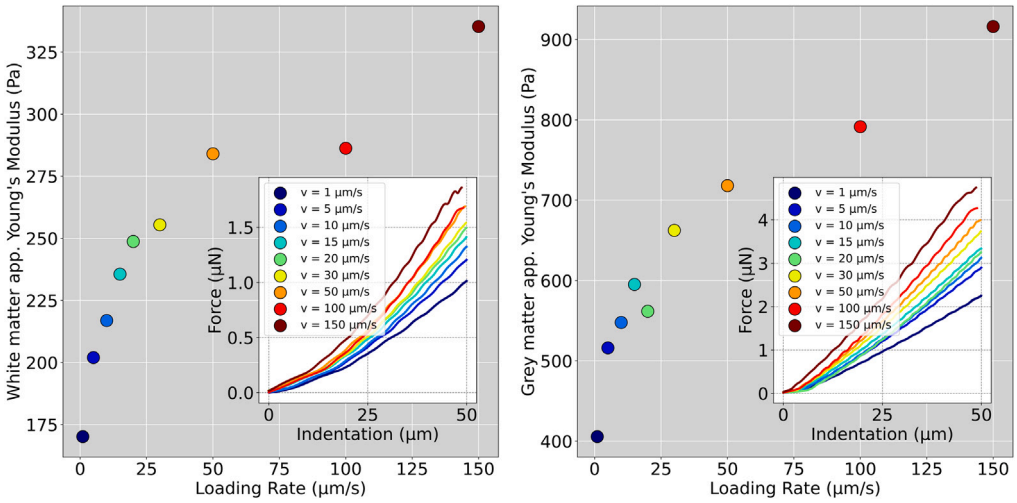


Fig. 8. Exemplary loading rate sensitivity of spinal cord grey and white matter tissue. The apparent modulus increases with increasing loading rate. The inset shows the corresponding loading phases. Experimental conditions: 18 indentations on two locations (nine indentations each) of one sample, maximum indentation depth of 50 µm, varying loading rate, holding time of 10 s, tip radius of 98 µm.

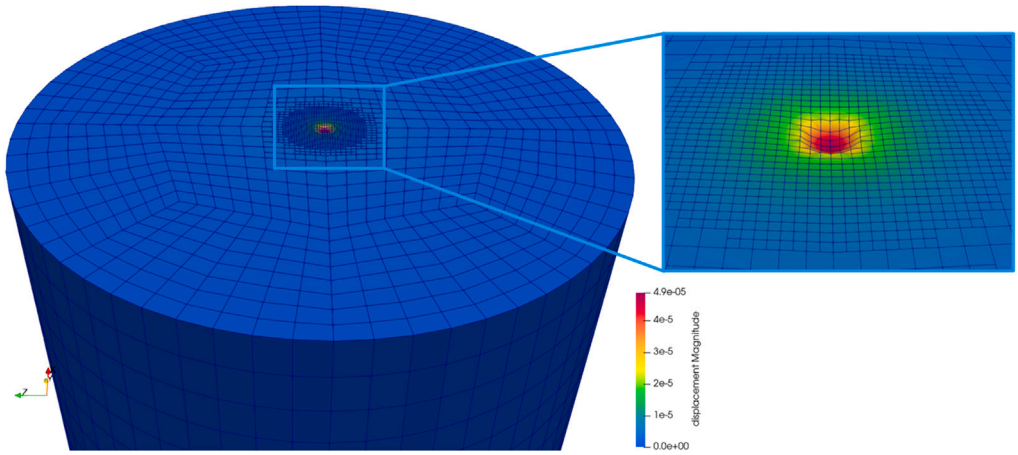


Fig. 9. Finite element simulation of an indentation experiment with 50 µm maximum depth and a spherical indenter with a radius of 100 µm. Geometrical and numerical parameters: 2.5 mm radius, 5 mm height, 0.05 mm maximum element size and 0.005 mm local element size.

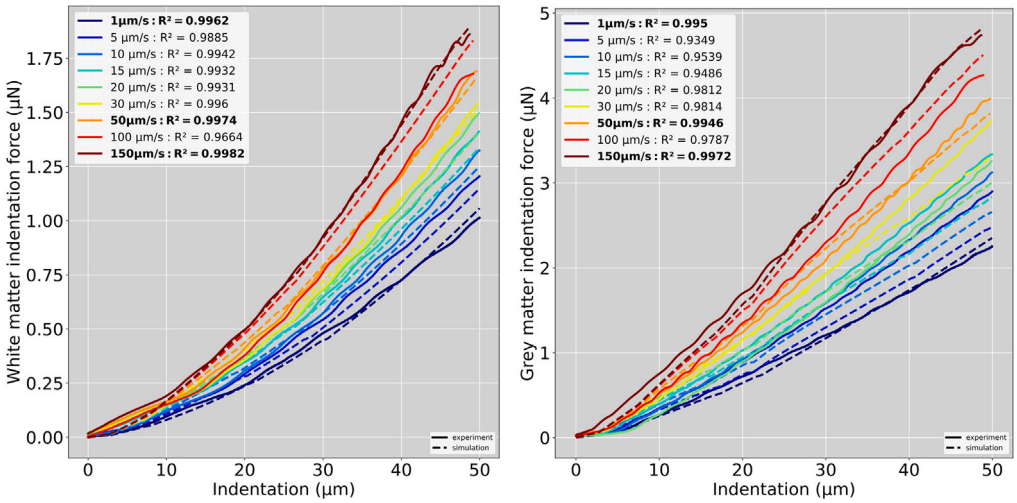


Fig. 10. Results of the inverse parameter identification for the viscoelastic Ogden model for spinal cord white (left) and grey (right) matter tissue. Comparison between experimental (solid lines) and simulated (dashed lines) force–displacement curves. The loading rates used during parameter identification are highlighted in bold in the legend.

Table 2
Optimized parameters for spinal cord grey and white matter tissue.

	μ_∞	α_∞	μ_1	α_1	η_1
Grey matter	128.75	7.85	295.29	9.13	79.02
White matter	31.78	−22.05	54.08	−29.25	40.34

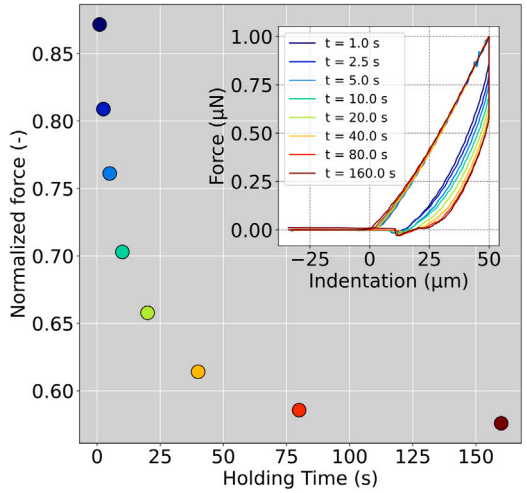


Fig. 11. Exemplary holding time sensitivity of spinal cord grey matter tissue. Dependence of the relaxed force on the holding time highlights the rate-sensitive characteristics of spinal cord tissue. As holding times increases, the force relaxation shows a corresponding increase. Simultaneously, the decrease in force converges with prolonged holding times towards its plateau value. Experimental conditions: eight indentations on one sample, maximum indentation depth of 50 μm, loading rate of 30 μm/s, varying holding time, tip radius of 98 μm.

The model for spinal cord grey matter tissue displayed on the right side of Fig. 10 also shows strong predictive capabilities, as evidenced by an overall $R^2 > 0.93$ for all loading rates. However, the experimental data show non-physical behavior for a loading rate of 20 μm/s, as it implies a softer mechanical material response than at a loading rate of 15 μm/s. This resulted in a relatively low R^2 for 15 μm/s. In addition, the simulated results for 5 μm/s show a lower R^2 for spinal cord grey and white matter tissue. Fig. 8 shows that the apparent moduli for both curves are also larger than the general trend, which could explain the low R^2 values.

3.5. Mechanical properties of spinal cord tissue under stress relaxation

To further investigate the viscoelastic nature of spinal cord tissue, we performed eight experiments on the same location of grey matter with different holding times of 1 s, 2.5 s, 5 s, 10 s, 20 s, 40 s, 80 s, and 160 s as well as a constant loading rate of 30 μm/s. In addition, we performed 12 experiments, six on spinal cord white and six on spinal cord grey matter tissue with an extended holding time of 600 s to assess the long-term stress relaxation behavior.

Fig. 11 displays increased relaxation and an asymptotically converging curve towards an equilibrium value with increasing holding time. In the range of 1 s to 160 s holding time, the normalized force decreased from 0.872 to 0.576, which is equivalent to a reduction by one third.

Fig. 12 illustrates the results of our experiments with an extended holding time of 600 s. Only the first 300 s are shown here as the curves had converged towards the equilibrium value at around 300 s. We employ the multi-term Prony series in Eq. (2) with n characteristic time constants τ_i to quantify the viscoelastic response of spinal cord tissue. The plateau stress σ_∞ at $t \to \infty$ is related to the storage modulus and represents the equilibrium elastic response; the difference between peak stress and plateau stress $[\sigma_i - \sigma_\infty]$ is related to the loss modulus and represents the viscoelastic response. The averaged curves of the three indentation experiments on spinal cord white or grey tissue, respectively, were used for the fitting. The best approximation of the experimental data was achieved with a two-term Prony series.

Table 3
Characteristic exemplary time constants τ_1 and τ_2 for spinal cord grey and white matter tissue determined by approximating the stress relaxation behavior by a two-term Prony series through a least square fit.

	τ_1	τ_2	R^2
Grey matter	1.38	36.29	0.9868
White matter	1.46	46.10	0.9837

Table 3 summarizes the corresponding exemplary time constants. Spinal cord white and grey matter tissue show very similar relaxation behavior with a fast time constant of τ_1 =1.46 s and 1.38 s, respectively, and a slow time constant of τ_2 = 46.10 s and 36.29 s, respectively. We conducted all subsequent experiments with a holding time of 10 s to minimize experimental and preservation time.

3.6. Conditioning behavior under cyclic loading

To test the response of spinal cord tissue under cyclic loading, we performed three cyclic indentation experiments on grey and three on white matter tissue, each with five cycles, a maximum indentation depth of 50 μm, and a loading and unloading rate of 10 μm/s. Fig. 13 illustrates the results of these experiments, where one of the three force–displacement curves on spinal cord grey matter tissue is shown on the left side and the relative peak forces and apparent moduli of the average curves for all five loading cycles on the right.

Our exemplary results show that the apparent modulus decreases with increasing loading cycle. For this exemplary sample, the influence of cyclic loading on the decrease in the apparent modulus is stronger for spinal cord grey matter tissue than for white matter tissue, with a decrease of approximately 20 and 14% for grey and white matter, respectively. This characteristic is equally displayed in the drop in peak force, with a decrease of 8.3% and 5.8% in grey and white matter tissue after five cycles, respectively. The influence of repeated loading on the measured properties is reduced with increasing number of loading cycles on both spinal cord grey and white matter tissue, with the first applied loading having the greatest impact on the apparent modulus. The relative drop in apparent modulus due to the first loading in relation to the total drop of the modulus after the applied five cycles is approximately 60% for white and 55% for spinal cord grey matter tissue, i.e., more than half of the stiffness loss (by means of the apparent modulus) is caused by the initial loading.

3.7. Dynamic mechanical analysis

Fig. 14 illustrates the results of our dynamic mechanical analysis (DMA) of porcine spinal cord tissue. Preliminary experiments with constant frequency and changing amplitudes showed no significant change in the mechanical properties. Therefore, for further investigations, we used a constant amplitude of 1 μm.

We conducted a series of 8 DMA tests with the frequencies 0.5, 1, 2, 4, 10 and 20 Hz on differing indentation depths of 10, 20, 30 and 40 μm for spinal cord grey and white matter tissue. It is clearly observable that for the two exemplary locations, the spinal cord grey matter depicts a lower overall storage modulus (E') and a similar loss modulus (E''), resulting in a lower overall $\tan(\delta)$ for spinal cord grey than for the spinal cord white matter tissue. The results are all derived by fitting the experimental data to Eq. (3).

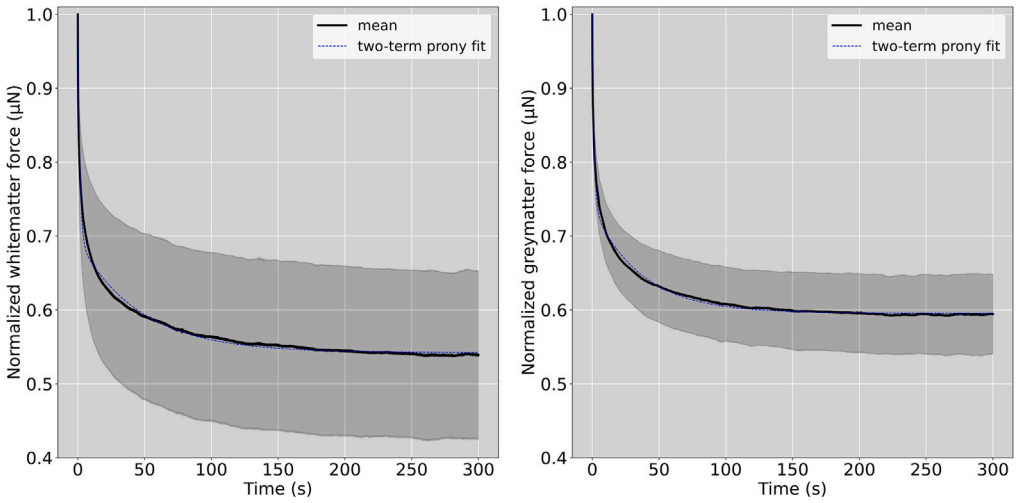


Fig. 12. Exemplary stress relaxation of spinal cord grey and white tissue. The plateau force is related to the storage modulus and describes the equilibrium elastic response. The discrepancy between peak force and plateau force is associated with the loss modulus and characterizes the viscoelastic response. Black lines represent the average response and grey areas show the standard deviation. Blue dashed lines show the performance of the two-term Prony series fit. Experimental conditions: 12 indentations on 12 different locations of one sample, maximum indentation depth of 50 µm, loading rate of 30 µm/s, holding time of 600 s, tip radius of 102 and 110 µm (each tip was used for six indents, three in white and three in grey matter).

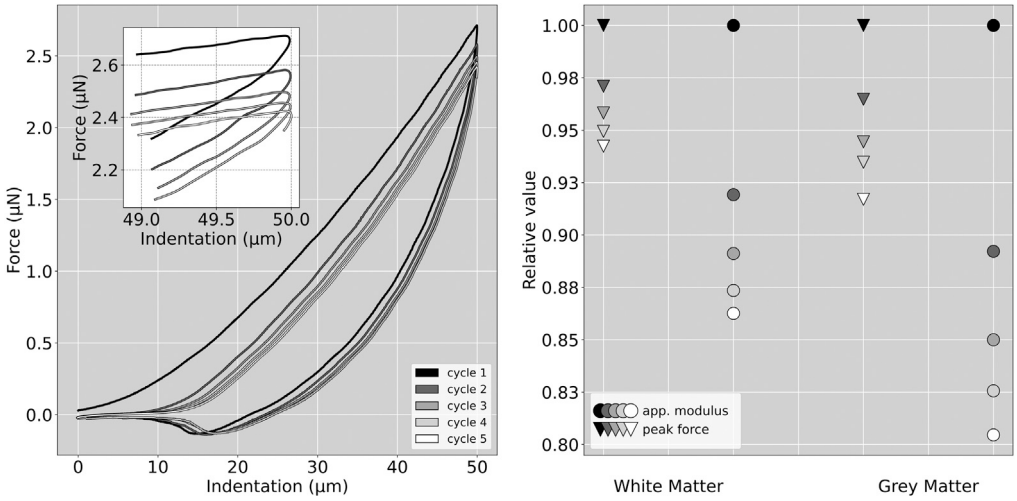


Fig. 13. Cyclic loading of spinal cord grey and white matter tissue. The left side illustrates the five characteristic loading/unloading force–displacement curves for an indentation experiments on grey matter. The right side shows the normalized decrease in maximum force and apparent modulus of the averaged curves of three experiments on spinal cord white and grey matter tissue respectively. Experimental conditions: five cycles of indentations on three locations in spinal cord grey and three locations in spinal cord white matter tissue on one sample, maximum indentation depth of 50 µm, loading and unloading rate of 10 µm/s, no holding time, tip radius of 110 µm, the values given are the mean values of three indentations in spinal cord grey matter and three indentations in spinal cord white matter.

3.8. Thermomechanical properties of spinal cord tissue

Fig. 15 shows the exemplary results of consecutive spherical indentations with a loading rate of $v = 50$ µm/s and varying sample temperature. The apparent moduli derived from these force-indentation curves clearly show that the stiffness depends on the temperature. The higher the temperature, the lower the apparent modulus. Simultaneously, the sample shrinks as it cools, as can be seen in Fig. S4. The observations in Fig. 15 highlight that regardless of the nature of thermal loading, heating up or cooling down, the general trend is that the tissue stiffness of porcine spinal cord measured by apparent modulus is inversely proportional to the temperature.

To further investigate the thermomechanical properties of spinal cord tissue, we performed a larger study with 549 indentations on 18 samples from three different porcine spinal cords. Here, we consecutively tested spinal cord grey matter and white matter tissue during heating up the sample from room temperature to body temperature. We then computed the trend of the average relative apparent modulus. The results of this study are shown in Fig. 16. We can clearly observe a similar behavior of softening for both spinal cord grey and white matter tissue as well as a stable trend of the ratio of grey matter tissue apparent moduli divided by white matter apparent moduli, as shown in the inset of Fig. 16.

3.9. Heterogeneity of spinal cord tissue properties

To visualize the local differences in spinal cord grey and white matter tissue stiffness, we conducted a grid scan of 1200 indentation points with 40 steps in horizontal and 30 steps in vertical direction with a step size of 300 µm covering a whole transverse plane of one exemplary spinal cord slice. We used a loading rate of 30 µm/s for this experiment. The results for the derived apparent modulus are shown in Fig. 17 together with an image of the corresponding spinal cord slice. The map of the apparent moduli highlights the typical shape of

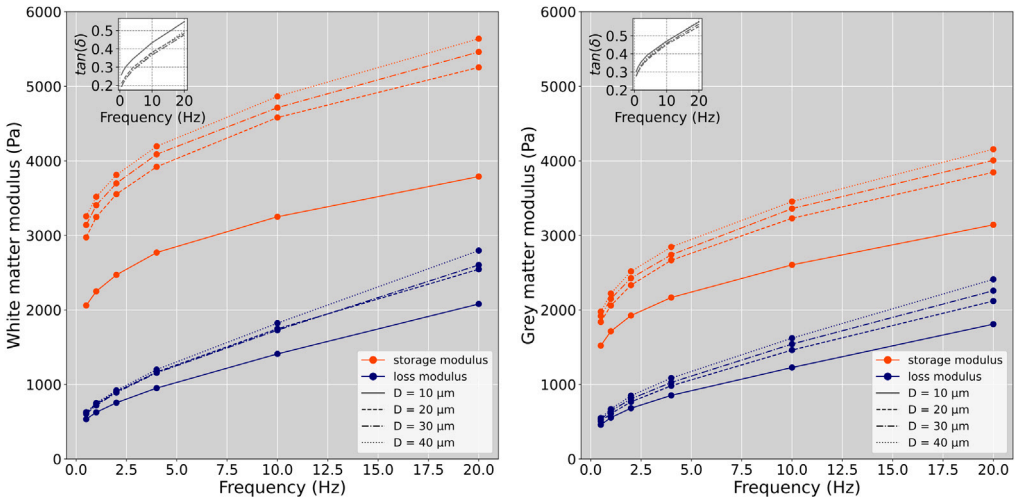


Fig. 14. Dynamic mechanical analysis (DMA) of spinal cord tissue. Left side shows the exemplary results for spinal cord white, right side for spinal cord grey matter tissue with the storage modulus E'' in orange and the loss modulus E' in blue. The ratio of loss modulus to storage modulus, $\tan(\delta)$ is shown in the inset at the top left. Experimental conditions: four DMA on two locations of one sample, varying maximum indentation depth, oscillatory amplitude of 1 µm, varying frequencies, tip radius of 105 µm.

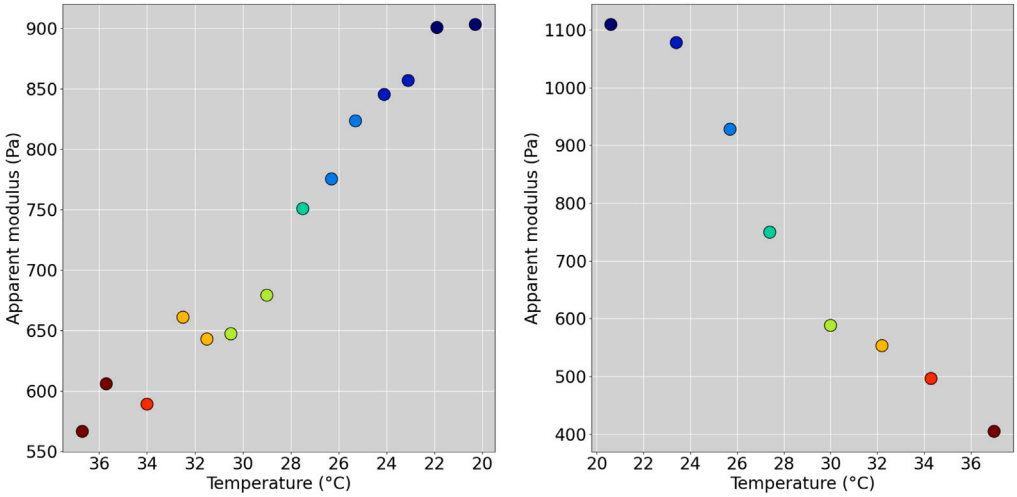


Fig. 15. Temperature sensitivity of spinal cord grey (left side) and white matter tissue (right side). The left and right side show the results for consecutive measurements during the process of decreasing and increasing the temperature, respectively. Experimental conditions: 14 and 8 indentations at decreasing and increasing temperature, respectively, on two samples from two porcine spinal cords, maximum indentation depth of 50 µm, loading rate of 50 µm/s, no holding time, tip radius of 98 µm.

grey matter in the spinal cord with higher values than the surrounding white matter. We also observe a small transitional phase regarding the mechanical properties between grey and white matter. From these results, we could compute exemplary average apparent moduli for spinal cord white and grey matter tissue for this particular sample with values of 128.7 and 403.8 Pa, respectively.

4. Discussion

Quantifying the tissue-scale mechanical properties of spinal cord tissue is challenging due to its geometry with an average diameter of 4.7 ± 3.5 to 13.3 ± 2.2 mm for human spinal cord (Frostell et al., 2016). This prevents testing grey and white matter properties individually with established macroscale experimental protocols (Budday et al., 2017a; Reiter et al., 2021; Hinrichsen et al., 2022; Faber et al., 2022). Therefore, we have developed a robust protocol to complement tissue-scale experiments that can only quantify the homogenized response of spinal cord tissue (containing both grey and white matter) through nanoindentation measurements. The established testing setup allows us to provide reliable and repeatable experimental results for spinal cord grey and white matter tissue individually. We conducted a series of indentation tests of freshly harvested porcine spinal cord tissue to investigate possible influences of different radii of spherical indenters, various loading rates, holding times, ambient temperatures and buffer submersion time as well as the local heterogeneity of spinal cord tissue. These investigations revealed physically explainable trends and consistent results, which form the basis for the development of testing protocols that will allow us to complement large-strain compression, tension, and shear experiments to calibrate nonlinear material models in the future (Faber et al., 2022).

4.1. Choosing the right indenter tip and experimental duration

The choice of the tip geometry depends to a large extent on the material to be tested. For testing materials such as metals or ceramics, three-sided pyramid-shaped tips with a sharp end are usually preferred, in contrast to indentation tests of biological tissues (Ebenstein and Pruitt, 2006). In the latter case, spherical indenters or flat punches are used to avoid tissue damage and stress concentrations due to the sharp end of pyramidal tips. Even with flat indenters, a high stress concentration can occur at the contact edge, which is why spherical indenters are most commonly used in biological tissue tests, such as

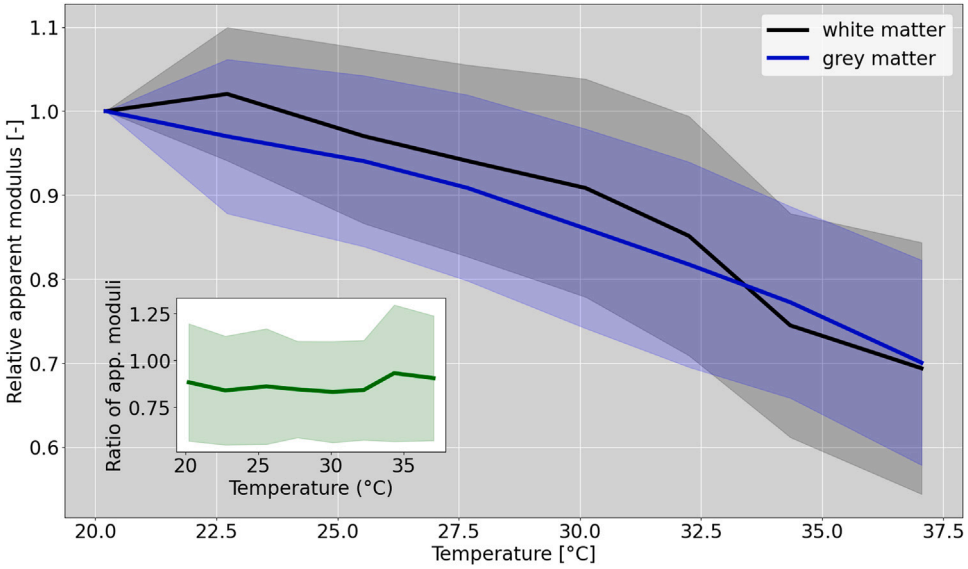


Fig. 16. Temperature sensitivity of ratio between spinal cord grey and white matter tissue stiffness. The blue and black line show the mean relative apparent moduli for white and grey spinal cord tissue and the corresponding standard deviation over the temperature ranging from room temperature (20 °C) to body temperature (37 °C). The inset plot shows the trend of the ratio of grey matter apparent moduli divided by white matter apparent moduli. Experimental conditions: 549 indentations on 18 samples from three porcine spinal cords, maximum indentation depth of 50 μm, loading rate of 50 μm/s, holding time of 1 s, tip radius of 97.5 μm.

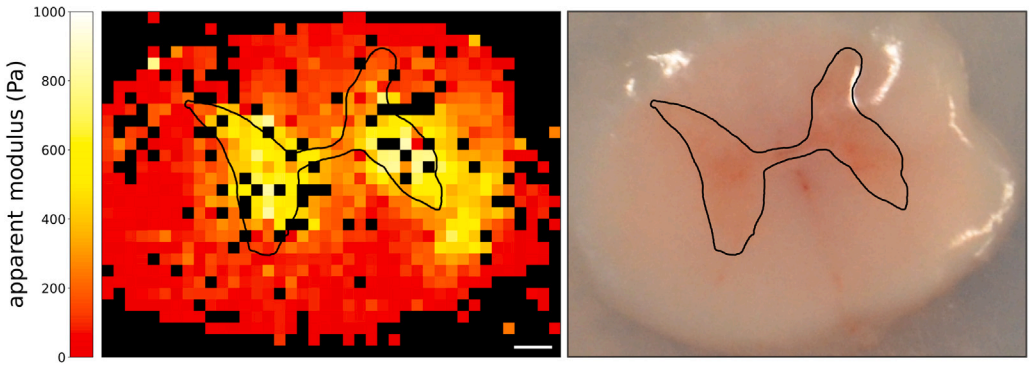


Fig. 17. Heat map of the apparent modulus of one exemplary spinal cord transverse plane with stiffer areas highlighting the characteristic shape of spinal cord grey matter tissue on the left side. Scale bar denotes 1 mm. Image of the corresponding spinal cord slice on the right side. Outlines of spinal cord grey matter tissue were added manually. Experimental conditions: 1200 indentations on one sample, maximum indentation depth of 50 μm, loading rate of 30 μm/s, holding time of 10 s, tip radius of 102 μm.

spinal cord indentation experiments (Gefen and Margulies, 2004; Gupta et al., 2007; Van Dommelen et al., 2010). During our measurements, it became clear that indentation tests with a spherical indenter with a radius of 100 μm provided the most consistent results compared to tests with indenters with a radius of 50 or 250 μm. Therefore, we performed all experiments except the radius comparison test with a 100 μm radius indenter. In addition, our initial studies have shown that the use of DPBS for indentation tests appears to be appropriate, as the general trend on average shows only minimal softening effects over time compared to experiments with distilled water as buffer. Nevertheless, the effects should be interpreted critically for experiments with an immersion time of more than one hour, as this could represent further softening trends due to tissue degradation over time.

4.2. Elastic properties of spinal cord grey and white matter tissue

The results of Fig. 17 show an average apparent modulus of 128.7 (n = 666) and 403.8 Pa (n = 149) for indentations with a maximum depth of 50 μm and a loading rate of 30 μm/s on spinal cord white and grey matter tissue, respectively. This indicates that spinal cord grey matter is stiffer than spinal cord white matter tissue, on average by a factor of 3.1 for this one sample. Notably, Fig. 17 shows a stiffer area in the spinal cord white matter tissue below the right ventral horn of the gray matter, which we are currently unable to explain. In future studies, we will incorporate histological analysis, which may help elucidate such observations. Our results are in strong agreement with the literature in the field of atomic force microscopy (AFM) on spinal cord tissue. Studies show that spinal cord grey matter tissue is ~1.8 times stiffer than spinal cord white matter tissue for adult zebrafish using force-controlled AFM experiments with $R = 18.65 \pm 0.15$ μm, a maximum force of 4 nN, and a loading rate of 10 μm/s (Möllmert et al., 2020), 2.6 times stiffer spinal cord grey matter tissue for juvenile mice for a similar setup with a maximum force of 7 nN (Koser et al., 2015), and 2.4 times stiffer spinal cord grey matter tissue for adult rat (Moeendarbary et al., 2017) with a spherical tip diameter of 89.3 μm, a maximum force of 20–30 nN (which resulted in an average 4 μm maximum indentation depth), and a loading rate between 5 and 10 μm/s. The AFM results for mammalian spinal cord thus fit well to the results obtained in the current study. The absolute values, however, are in general approximately 85% smaller for a tip radius of 18.65 ± 0.15 μm and 69% smaller for a tip radius of 44.65 μm as compared to the results obtained here. This can be attributed to the lower maximum indentation depth in Moeendarbary et al. (2017), where it is on average an order of magnitude smaller than the maximum indentation depth

of 50 μm used here. In the other two studies (Möllmert et al., 2020; Koser et al., 2015), even lower maximum forces were exerted on the spinal cord tissue, suggesting a lower maximum indentation depth, thus resulting in a lower apparent modulus compared to our measurements due to the nonlinearity of the tissue response. In this context, it should be noted that the results of macroscopic indentation tests on brain tissue indicate the opposite trend than our findings, i.e., the white matter in bovine brain (Budday et al., 2015) and pig brain (Kaster et al., 2011; Van Dommelen et al., 2010) is 36%–50% stiffer than the grey matter.

4.3. Viscolastic properties of spinal cord grey and white matter tissue

Figs. 7 to 12 reveal the highly rate-sensitive material behavior of the investigated spinal cord sample. With an increase in loading rate from 1 to 150 μm/s, both approximately doubled their peak forces and corresponding apparent moduli. This is in agreement with flat punch indentation experiments on bovine brain tissue in Budday et al. (2015). In absolute terms, spinal cord grey matter tissue shows a slightly higher sensitivity to the applied loading rate with an increase in peak force by a factor of 2.1 and an increase in apparent modulus by a factor of 2.3 compared to increases by factors of 1.8 and 2.1, respectively, for spinal cord white matter tissue. These results are consistent with the previously reported increase in spinal cord tissue modulus when the loading rate is increased (Jin et al., 2022, 2023). Apart from the two studies mentioned above, only Hung et al. (1981) carried out compression tests on the spinal cord of cats and found no significant influence of the stress rate or repeated stress, which could be attributed issues related to tissue preparation and the choice of experimental setup.

To quantify the rate-dependent response of the spinal cord grey and white matter tissue of the tested sample with a single set of parameters instead of providing individual apparent moduli for each loading rate, we ran finite element simulations of the indentation loading procedure to inversely identify parameters for a finite viscoelastic Ogden model with one viscoelastic element (Fig. 4). The model well predicts the increasing slope of the force-indentation curves for increasing loading speed, although only three of the nine loading speeds were used for the model calibration. The identified parameters, summarized in Table 2, yield higher shear moduli for spinal cord grey than white matter, while the absolute values of the nonlinearity are higher for white matter. The viscosity is slightly higher for spinal cord grey matter than for white matter. Compared to the results of Hinrichsen et al. (2022), especially the values for the nonlinearity parameter α, which also accounts for the typical compression-tension asymmetry observed for brain tissue, differ quite drastically. While Hinrichsen et al. (2022) consistently show negative values for α, we observed the opposite for the results of spinal cord grey matter tissue. This can be attributed to the limited information of indentation data related to the compression-tension asymmetry of the tissue, as mostly compression is applied in our nanoindentation experiments. This emphasizes that more experimental data, e.g., also including large-stain compression and tension loadings, will be needed in the future to appropriately calibrate mechanical models for spinal cord tissue.

The stress relaxation experiments showed a comparable response for spinal cord grey and white matter tissue of the tested sample with a force relaxation of 54 and 59%, respectively, within 300 s of holding. The stress relaxation of spinal cord white matter tissue was slightly more pronounced than for grey matter. The only other study on indentation-based stress relaxation of spinal cord tissue employed a spherical indenter with a radius of 250 μm, a loading rate of 40 μm/s, and a maximum indentation depth of 250 μm which was held for up to 30 s (Jin et al., 2022). With the holding time of 30 s used in the aforementioned study, it can be observed that equilibrium was not reached, which agrees with our results. In their study, however, the relaxation curve was not analyzed with a Prony fit. It is also worth noting that, unlike our experiments on transverse spinal cord slices, they performed indentations on the outermost sagittal plane of an entire spinal cord segment (i.e., white matter) and did not report the removal of the innermost meninges of the spinal cord segment, which has been shown to have a stiffening effect on spinal cord tissue during tensile testing (Ramo et al., 2018b). Compared to our final relaxation force, their results show a decrease in peak force of 46% after the applied hold time of 30 s. This differs by more than 10% from our results for spinal cord white matter relaxation force after 300 s and emphasizes the pronounced relaxation time of the spinal cord tissue.

Other related experiments investigating the viscoelasticity of spinal cord tissue under tension were conducted on lower thoracic dog (Tunturi, 1980), first and second lumbar cat Chang et al. (1988), cervical and thoracic human (Bilston and Thibault, 1995) and first cervical to fourth lumbar rat spinal cord tissue (Fiford and Bilston, 2005). Tunturi (1980) performed quasi-static experiments, while the remaining three used strain rates of 0.003 to 0.0012 s^{-1}, 0.048 to 0.225 s^{-1} and 0.002 to 0.2 s^{-1}, showing time periods until full equilibrium of 15 min, approximately 2 min, below 25 s and approximately 8–15 min, respectively. This wide range of results can most likely be attributed to the large differences in strain rates, preconditioning, *post mortem* time, species, sample size, and testing modality. These results illustrate the complexity of the experimental investigation of viscoelastic material properties of spinal cord tissue and thus the ambiguity when comparing mechanical material parameters.

To further quantify the stress relaxation behavior, we fitted a Prony series to the averaged relaxation curves of spinal cord grey and white tissue. For our data, we found that a two-term Prony series with two time constants τ_1 and τ_2 provides the best approximation, capturing both the initial and late relaxation phases. The time constants of spinal cord grey and white matter tissue are similar, with slightly lower values for grey matter. For both regions of spinal cord tissue, τ_1 and τ_2 differ by an order of magnitude. This might be attributed to one time constant being associated with the viscous and one with the porous nature of spinal cord tissue. Interestingly, a very similar viscoelastic behavior was observed for brain tissue. Budday et al. (2015), Gefen and Margulies (2004) and Prange and Margulies (2002). The only studies that had previously fit spinal cord tissue stress relaxation data to a viscoelastic model used tensile test data and employed a four-term Prony series with fixed time constants at $\tau_1 = 0.1$, $\tau_2 = 1$, $\tau_3 = 10$ and $\tau_4 = 100$ s (Ramo et al., 2018b; Shetye et al., 2014; Ramo et al., 2018a). This approach made it difficult to compare with our results, as they prefixed the relaxation time constants before fitting their model with the stress relaxation data.

The effects of preconditioning on the mechanical properties of spinal cord tissue observed in our cyclic loading experiments further highlight its viscoelastic nature. The observed softening effects of cyclic preloading are consistent with reported observations in the literature (Clarke, 2010). Cheng et al. (2009) measured the softening effect of preconditioning during uniaxial tests on spinal cord tissue of rats. The experiments were performed on three different groups with 5%, 5 % and 2% preconditioning strain and 5%, 2% and 2% testing strain. They reported that the peak stress of the third group was 134% higher than that of the other two groups, which is consistent with the softening tendency of preconditioning that we observed. Ramo et al. (2018b) reported a drop in peak stress during dynamic cyclic tensile testing with a deformation of 5% strain, after removing the construct of the innermost meninges, by 31.5 ± 8.1, and 39.9 ± 9.7% for cyclic loading frequencies of 0.16 and 0.016 Hz respectively. Our results, which were obtained at a loading frequency of 0.1 Hz, show this softening behavior to a greater extent, which may be due to the higher local deformation gradient during our nanoindentation tests as compared to 5% tensional strain. A similar decrease in stiffness for cyclic uniaxial compression-tension tests on human brain tissue was observed in Franceschini et al. (2006). Although we could not find any studies on cyclic testing using

indentation, the experimental efforts mentioned above all show the softening effect of preloading, regardless of the test setup.

The dynamical mechanical analysis (DMA) of the tested spinal cord tissue sample reveals the typical increase in storage and loss modulus with increasing oscillatory frequency, as previously observed for central nervous system tissue (Hrapko et al., 2006; Shen et al., 2006; Nicolle et al., 2005). In comparison, spinal cord grey and white matter tissue show a very similar energy loss due to friction and internal motions (Menard and Menard, 2020), represented by the viscous loss modulus E''. In contrast, spinal cord grey matter tissue shows a lower elastic modulus E' compared to white matter tissue. Notably, this trend is opposite to what we observed during our quasi-static experiments and once again highlights that the stiffness differences between grey and white matter highly depend on the loading conditions due to the poro-viscoelastic nature of the spinal cord tissue response. In addition, it is worth noting that the loading scenario during the DMA is characterized by small deformations with an amplitude of 1 μm, which describes a fundamentally different loading situation than the quasi-static experiments, which are tests with large deformations. This might also contribute to the opposing results. There are few studies that we could find on compressive indentation-based DMA on spinal cord tissue. The only related study performed macroscopic compression on the sagittal outer plane of a whole spinal cord segment after removing the dura mater from the sample with an impactor of 5 mm diameter (Fradet et al., 2016). They reported no significant change in loss or storage modulus with 34, 49 and 39 Pa and 42, 48 and 44 Pa at frequencies of 1, 10 and 20 Hz, respectively. These results seem to be rather low compared to our experiments, but considering the testing machine used and the measured stresses in the MPa range, the comparability with our study is questionable. Furthermore, we could show that within our frequency spectrum at indentation depths 20, 30 and 40 μm, the dynamic moduli seem to change insignificantly compared with the difference to the experiments at 10 μm indentation depth for both spinal cord grey and white matter tissue. This indicates that a prestrain of up to 20 μm appears to have a major influence on the complex modulus determined by dynamic mechanical analysis.

4.4. Thermoelastic properties of spinal cord tissue

Interestingly, we observed a significant increase in spinal cord tissue stiffness (derived from indentation loading curves and the Hertzian contact theory) with decreasing temperature (Fig. 15). The apparent modulus decreased by approximately 30% for both spinal cord grey and white matter tissue for increasing temperature from 20 °C to 37 °C (Fig. 16). The same temperature-dependent properties were observed when decreasing the temperature of the sample. Interestingly, the ratio of spinal cord grey matter to white matter tissue stays approximately constant over the investigated temperature spectrum. This suggests that while the testing temperature is critical when determining absolute stiffness values for spinal cord tissue from indentation experiments, it is less decisive for the ratio between grey and white matter stiffness – as long as the temperature is the same for both types of tissue. Apart from this temperature sensitivity study and the investigation of the influence of the immersion time in buffer on the mechanical properties of spinal cord tissue, we conducted all experiments at a (room) temperature of approximately 20° to 22 °C, which is comparable to the temperatures of 18 and 20 °C reported in Möllmert et al. (2020) and Schlüßler et al. (2018).

4.5. Limitations

It is important to emphasize that the results of this study show mechanical characteristics of porcine spinal cord tissue under different loading scenarios based on indentations with a focus on qualitative observations. The number of animals and samples per test (with the exception of the temperature sensitivity experiments and the buffer submersion time sensitivity tests) is not sufficient to provide quantitative values for the mechanical properties of the tissue studied. Furthermore, this study does not investigate whether the mechanical properties vary at different levels of the spinal cord. Due to the harvesting procedure of the organs at the slaughterhouse, we could not expect to obtain the entire spinal cord intact, which would have allowed us to assign the samples to the cervical, thoracic or lumbar section of the porcine spinal cord. We will have to adapt this procedure to investigate region-dependent mechanical properties of porcine spinal cord tissue in a future study. In addition, we exclusively focus on the mechanical characteristics of spinal cord tissue by nanoindentation. This *ex vivo* test method requires a series of preparatory steps that could potentially alter the microstructure and the mechanics of the tissue. Therefore, there is reason to believe that there are discrepancies between the mechanical properties of spinal cord tissue in living organisms and the properties measured in this study. In addition, indentation measurements lead to a very local and inhomogeneous deformation state. A multi-modality experimental pipeline that includes *ex vivo* tests, such as large-strain cyclic loading with a rheometer (Budday et al., 2017a) in combination with nanoindentation and atomic force microscopy (Moeendarbary et al., 2017) at the meso-/ microscale, as well as *in vivo* experiments, such as magnetic resonance elastography (Sack, 2023) and Brillouin microscopy (Schlüßler et al., 2018), would enable a more comprehensive understanding of the mechanical material properties of *in* and *ex vivo* spinal cord tissue.

5. Concluding remarks

We have developed a robust, reliable, and repeatable protocol to characterize the mechanical properties of spinal cord grey and white matter tissue. All indentation tests were performed on freshly harvested porcine spinal cord tissue by the use of a Chiaro Nanoindenter™. We used spherical tips, which is commonly done for indentation experiments on biological tissue. We were able to obtain consistent, physically interpretable results indicating that spinal cord grey matter tissue is stiffer than white matter tissue. Our tests showed that the ambient temperature, the buffer submersion time, and the applied loading rate can significantly change the derived apparent modulus of spinal cord tissue. The viscosity of spinal cord white and grey matter tissue seems to be similar, with comparable relaxation times. In agreement with these findings, the dynamic mechanical analysis revealed similar loss moduli for both spinal cord grey and white matter tissue and higher storage moduli for white matter. Combining the presented mechanical characterization approaches with further testing modalities (*ex vivo* and *in vivo*) will make the complex mechanical nature of spinal cord tissue accessible. This will help to integrate in silico simulations to develop new treatments for spinal cord injury and diseases.

CRediT authorship contribution statement

Oskar Neumann: Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Harsh Vardhan Surana:** Investigation. **Stephen Melly:** Writing – review & editing, Software, Methodology, Investigation. **Paul Steinmann:** Writing – review & editing, Supervision, Funding acquisition. **Silvia Budday:** Writing – review & editing, Supervision, Project administration, Methodology, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary material related to this article can be found online at https://doi.org/10.1016/j.jmbbm.2024.106863.

Data availability

Data will be made available on request.

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